File permissions in Linux

Project description:

The research team at my organization needs to update the file permissions for certain files and directories within the projects directory. The permissions do not currently reflect the level of authorization that should be given. Checking and updating these permissions will help keep their system secure. To complete this task, I performed the following tasks:

Check file and directory details

The following code demonstrates how I used Linux commands to determine the existing permissions set for a specific directory in the file system.

```
researcher2@8fd5d20f9c24:~$ pwd
/home/researcher2
researcher2@8fd5d20f9c24:~$ ls
projects
researcher2@8fd5d20f9c24:~$ cd projects
researcher2@8fd5d20f9c24:~/projects$ ls -a
.   .project_x.txt  project_k.txt  project_r.txt
..  drafts          project_m.txt  project_t.txt
researcher2@8fd5d20f9c24:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Apr 22 18:42 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Apr 22 18:42 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Apr 22 18:42 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 22 18:42 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 22 18:42 project_t.txt
researcher2@8fd5d20f9c24:~/projects$ ls -al
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 22 18:42 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 22 19:21 ..
-rw--w---- 1 researcher2 research_team   46 Apr 22 18:42 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Apr 22 18:42 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Apr 22 18:42 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Apr 22 18:42 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 22 18:42 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 22 18:42 project_t.txt
researcher2@8fd5d20f9c24:~/projects$ 
```

- Here I used multiple commands for checking the file and directory details . at first I used ls -a command to display the hidden file . then I used ls -l for Displays permissions to files and directories. At last I used **`ls -la`**: for Displays permissions to files and directories, including hidden files. This is a combination of the other two options.

Describe the permissions string

The 10-character string can be deconstructed to determine who is authorized to access the file and their specific permissions. The characters and what they represent are as follows:

- 1st character: This character is either a d or hyphen (-) and indicates the file type. If it's a d, it's a directory. If it's a hyphen (-), it's a regular file.

- 2nd-4th characters: These characters indicate the read (r), write (w), and execute (x) permissions for the user. When one of these characters is a hyphen (-) instead, it indicates that this permission is not granted to the user.
- 5th-7th characters: These characters indicate the read (r), write (w), and execute (x) permissions for the group. When one of these characters is a hyphen (-) instead, it indicates that this permission is not granted for the group.
- 8th-10th characters: These characters indicate the read (r), write (w), and execute (x) permissions for other. This owner type consists of all other users on the system apart from the user and the group. When one of these characters is a hyphen (-) instead, that indicates that this permission is not granted for other.

For example, the file permissions for project_t.txt are -rw-rw-r--. Since the first character is a hyphen (-), this indicates that project_t.txt is a file, not a directory. The second, fifth, and eighth characters are all r, which indicates that user, group, and other all have
read permissions. The third and sixth characters are w, which indicates that only the user and group have write permissions. No one has execute permissions for project_t.txt.

Change file permissions

The organization does not allow others to have write access to any files. files. To comply with this, I referred to the file permissions that I previously returned. I determined project_k.txt must have the write access removed for other.

```
researcher2@8fd5d20f9c24:~/projects$ chmod o-w project_k.txt
researcher2@8fd5d20f9c24:~/projects$ ls -al
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 22 18:42 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 22 19:21 ..
-rw--w---- 1 researcher2 research_team   46 Apr 22 18:42 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Apr 22 18:42 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Apr 22 18:42 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Apr 22 18:42 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 22 18:42 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 22 18:42 project_t.txt
```

so I used the o-w project_k.txt command to change the file permisson so that otheres wont be able to write access to any file and the ls -al command shows that no others has the write permission any file .

Change file permissions on a hidden file

The research team at my organization recently archived project_x.txt. They do not want anyone to have write access to this project, but the user and group should have read access. The following code demonstrates how I used Linux commands to change the permissions:

```
researcher2@8fd5d20f9c24:~/projects$ chmod u-w,g-w .project_x.txt
researcher2@8fd5d20f9c24:~/projects$ chmod g+r .project_x.txt
researcher2@8fd5d20f9c24:~/projects$ ls -al
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 22 18:42 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 22 19:21 ..
-r--r----- 1 researcher2 research_team   46 Apr 22 18:42 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Apr 22 18:42 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Apr 22 18:42 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Apr 22 18:42 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 22 18:42 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 22 18:42 project_t.txt
researcher2@8fd5d20f9c24:~/projects$
```

I know .project_x.txt is a hidden file because
it starts with a period (.). In this example, I removed write permissions from the user and group, and added read permissions to the group. I removed write permissions from the user with u-w. Then, I removed write permissions from the group with g-w, and added read permissions to the group with g+r.

Change directory permissions

My organization only wants the researcher2 user to have access to the drafts directory and its contents. This means that no one other than researcher2 should have execute permissions.
The following code demonstrates how I used Linux commands to change the permissions:

```
researcher2@8fd5d20f9c24:~/projects$ ls -al
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Apr 22 18:42 .
drwxr-xr-x 3 researcher2 research_team 4096 Apr 22 19:21 ..
-r--r----- 1 researcher2 research_team   46 Apr 22 18:42 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Apr 22 18:42 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Apr 22 18:42 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Apr 22 18:42 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 22 18:42 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Apr 22 18:42 project_t.txt
researcher2@8fd5d20f9c24:~/projects$
```

The output here displays the permission listing for several files and directories. Line 1 indicates the current directory (projects), and line 2 indicates the parent directory (home). Line 3 indicates a regular file titled .project_x.txt. Line 4 is the directory (drafts) with restricted permissions. Here you can see that only researcher2 has execute permissions. It was

previously determined that the group had execute permissions, so I used the chmod command
to remove them. The researcher2 user already had execute permissions, so they did not need to be added.

Summary

I changed multiple permissions to match the level of authorization my organization wanted for files and directories in the projects directory. The first step in this was using ls -la to check the permissions for the directory. This informed my decisions in the following steps. I then used the chmod command multiple times to change the permissions on files and directories.